Multi-MNO

Considerations for Multi-MNO data pipelines

Index

I. Introduction

Overview - learning objectives

This session explores the concept of combining data from multiple Mobile Network Operators (MNOs) to generate more accurate and representative indicators. You will learn about key challenges, methodologies, privacy concerns, and ongoing global initiatives:

- Compare the strengths and limitations of single vs. multi-MNO data strategies
- Identify and differentiate between key multi-MNO integration approaches
- Recognize privacy, legal, and operational challenges in multi-MNO environments
- Explore key global initiatives and case studies that illustrate the practical implementation of multi-MNO

Contents

- **II. Single MNO vs Multi-MNO**. This section compares the strengths and limitations of using data from a single mobile network operator versus combining data from multiple MNOs, focusing on aspects such as representativeness, bias, and data completeness.
- **III. Multi-MNO Approaches**. We explore various methodological approaches for integrating data from multiple MNOs, including both aggregated data sharing models (e.g., MOAC) and disaggregated privacy-preserving computation frameworks (e.g., MODEC).
- **IV. Multi-MNO Initiatives**. This part reviews key international initiatives that have implemented or experimented with multi-MNO strategies to produce high-quality official statistics, population insights, or mobility indicators.
- **V. Session Exercise**. A practical exercise is provided to help readers understand the implications of using single-MNO versus different types of multi-MNO approaches, through a simplified but illustrative example.
- **Annex I. Exercise Answer**. Contains the complete solution to the session exercise, explaining the expected outputs and reasoning behind each scenario.
- **Annex II. Further Reading**. Offers a curated list of documents for those who want to explore the multi-MNO landscape in greater depth.

II. Single MNO vs Multi-MNO

Single MNO vs Multi-MNO

Mobile Phone Data offers an unprecedented source of near-real-time, large-scale information about human mobility and presence patterns. Unlike traditional surveys or censuses, mobile data is continuously collected from millions of devices, covering broad populations and diverse geographic areas at low cost.

Data from a **single MNO** is valuable on its own, providing rich insights into user behavior and mobility patterns. However, when relying solely on one operator, certain limitations should be taken into account:

- **Partial market coverage**: even a large operator typically covers only a portion of the total population, which may affect representativeness.
- **User profile bias**: each MNO's subscriber base can differ demographically, for example, skewing younger or older, which influences the data characteristics.
- **Geographical bias**: network coverage and service quality may vary across regions, impacting data completeness and accuracy in some areas.

Single MNO vs Multi-MNO

By integrating data from **multiple MNOs**, it is possible to increase population coverage, reduce bias, and improve the reliability and accuracy of statistical indicators. However, this multi-MNO approach introduces several challenges that must be carefully managed:

- **Data compatibility**: Datasets from different MNOs must be comparable in terms of structure, event frequency, granularity, and quality. Inconsistencies can lead to distortions in the aggregated results.
- **Methodological consistency**: It is essential to apply the same definitions, filtering rules, and computational methods across all datasets to ensure harmonized outputs and avoid systematic discrepancies between operators. However, since it is very likely that the characteristics of the data are not exactly the same across all MNOs (e.g., one operator may record events more frequently than another), the methodology must be flexible enough to adapt to and take these differences into account. Additionally, special attention must be paid to SIMs that appear across multiple operators, as these can lead to double-counting if not properly deduplicated.
- **Privacy and legal compliance**: All data handling and sharing activities must strictly comply with data protection laws and privacy regulations.

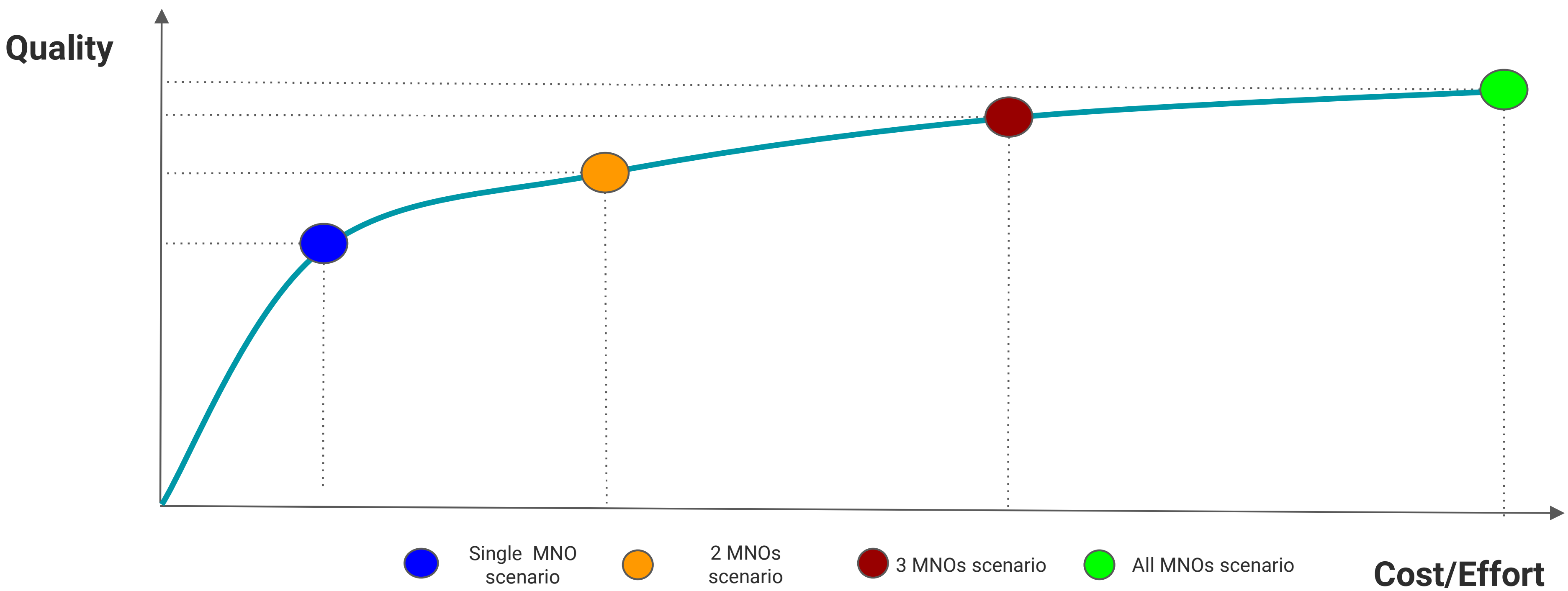
Single MNO vs Multi-MNO

- **Data access agreements:** Reaching agreements with MNOs to access data is often complex. The data is sensitive, not only from a personal privacy perspective but also from a business confidentiality standpoint. As a result, the willingness and modalities of collaboration may differ from one MNO to another.

- **Costs:** Incorporating additional MNOs typically involves higher operational costs, including increased demands on human resources, coordination efforts, and computational infrastructure. Therefore, it is important to evaluate the trade-offs of including more operators depending on the specific objectives of the analysis. In some cases, working with data from a single MNO may be more practical and still sufficient for the intended use case.

Integrating multiple MNOs can significantly enhance the quality of statistical outputs, but it also brings important technical, legal, and logistical challenges. A careful balance must be struck between improved accuracy and the feasibility of implementation, aiming for a solution that is both statistically robust and operationally sustainable.

Single MNO vs Multi-MNO



III. Multi-MNO approaches

Multi-MNO approaches

When working with mobile network data from multiple operators, one of the most critical decisions is how to combine the data in a way that maximizes statistical value while minimizing privacy risks and operational complexity. Multi-MNO integration is not a single method, but rather a **design choice** that requires balancing **data granularity**, **legal compliance**, and **technical feasibility**.

There are fundamentally different strategies for bringing together data from separate MNOs. These strategies vary according to how the data is structured (e.g., aggregated or device-level), what is shared between parties, and what kind of coordination or secure infrastructure is required. In all cases, the core challenge is to ensure that the combination of data leads to greater representativeness and precision, not simply a larger volume of information.

In some contexts, legal and privacy requirements limit the type of data that can leave an operator's environment. In others, advanced privacy-preserving techniques can enable more detailed joint computation.

Ultimately, choosing the right multi-MNO approach requires careful consideration of the privacy implications, technical constraints, and desired statistical outputs, as well as trust and governance agreements among the involved operators.

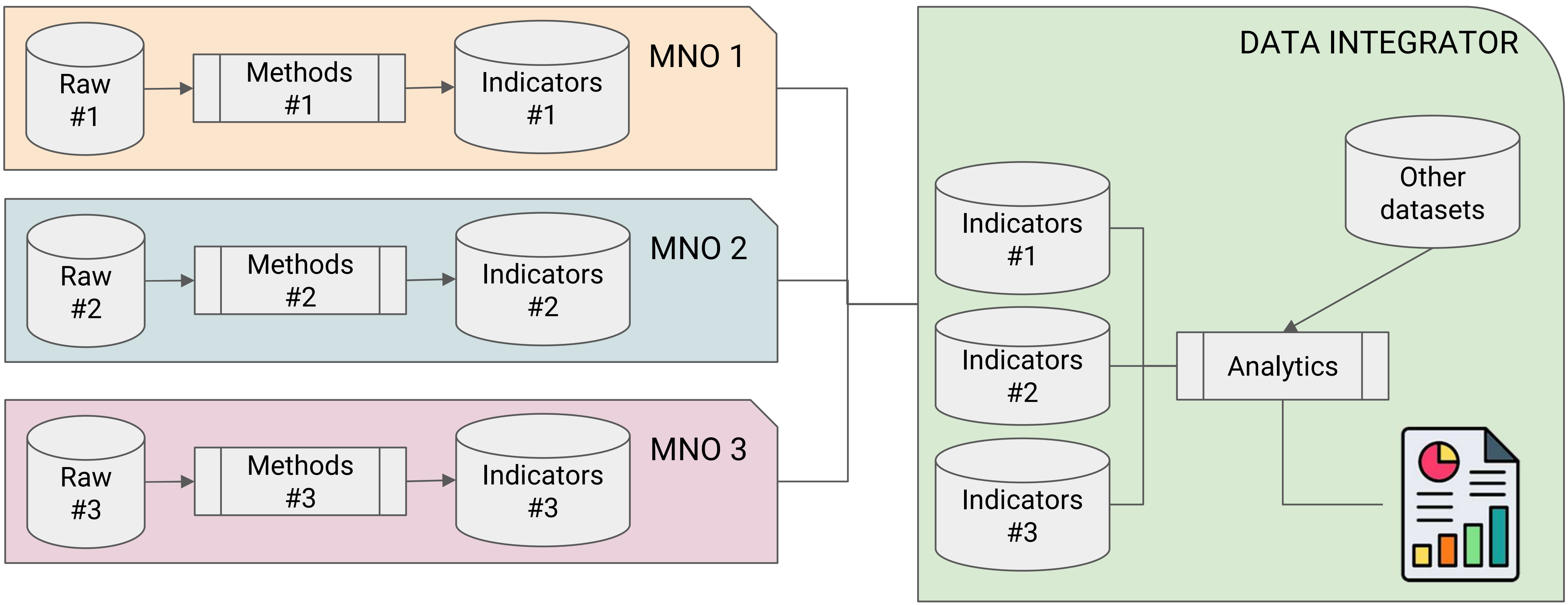
Multi-MNO approaches

In general terms, Multi-MNO approaches can be divided into two main groups:

- **Multi-Operator Aggregate-level Combination (MOAC)**: each MNO produces and shares aggregated indicators (e.g., number of devices per area and hour). These aggregates are combined statistically to build unified metrics. This approach is simple, privacy-safe, and easy to implement, but it may lead to double-counting and offers limited accuracy for deduplicated population estimates.

- **Multi-Operator DEvice-level Combination (MODEC)**: each MNO participates in a joint computation using Privacy-Enhancing Technologies (PETs). Pseudonymized device-level data is processed securely to estimate unique users across networks without revealing identities. This approach provides higher accuracy and deduplication, but requires greater coordination, technical infrastructure, and robust legal frameworks.

MOAC approach - general schema



MOAC approach

Phase 1: Internal MNO Data Handling

Each Mobile Network Operator processes its own raw network data independently. This data typically includes:

- Network events (e.g., location updates, handovers, call initiations)
- Location data (e.g., cell tower IDs or coordinates)
- Device identifiers (e.g., hashed or pseudonymized IDs)
- Occasionally, contract socio-demographic information (e.g. age, gender)

MOAC approach

Phase 1: Internal MNO Data Handling

This raw data remains internal to each MNO and is never shared directly. Instead, each operator usually performs the following local processing:

- **Data filtering and pre-processing**: cleaning (syntactic and semantic) and selecting relevant events based on, for example, event type (e.g. roaming-in event), time windows or region.
- **Device-level information**: applying rules to define user presence, mobility, or other behaviors based on internal methodology.
- **Aggregation**: computing totals per spatial and temporal unit (e.g., number of devices per 1 km^2 per hour).
- **Privacy protection**: before any data is shared externally, privacy thresholds are enforced. Most commonly k-anonymity, where aggregated data is only shared if it includes at least *k* devices (e.g., $k \geq 10$).

MOAC approach

Phase 1: Internal MNO Data Handling

Once processed, only anonymous, aggregated indicators are shared with an external third party who plays the role of data integrator (e.g. National Statistical Institute).

IMPORTANT: each MNO may apply different methodologies, privacy thresholds, and formats, which introduces variability in the final combined output.

MOAC approach

Phase 2: Integration and Output

The Data Integrator is a trusted third party responsible for merging the aggregated indicators from multiple MNOs into coherent, unified statistics. It does not access raw or personal data.

Main functions include:

- **Quality Control and Normalization**: Validating data structure, formats, and resolution, and ensuring comparability across operators.
- **Statistical Fusion**: Combining indicators through methods such as:
    - Weighted sums (e.g. adjusting for MNO market share)
    - Calibration using external sources (e.g., census data)
    - Corrections for double-counting (when the same user appears in more than one MNO dataset)

MOAC approach

Phase 2: Integration and Output

- **Output Generation**: Creating indicators like:
    - Presence by area and hour
    - Flows between zones or regions
    - Duration of stay and mobility rates

MOAC approach

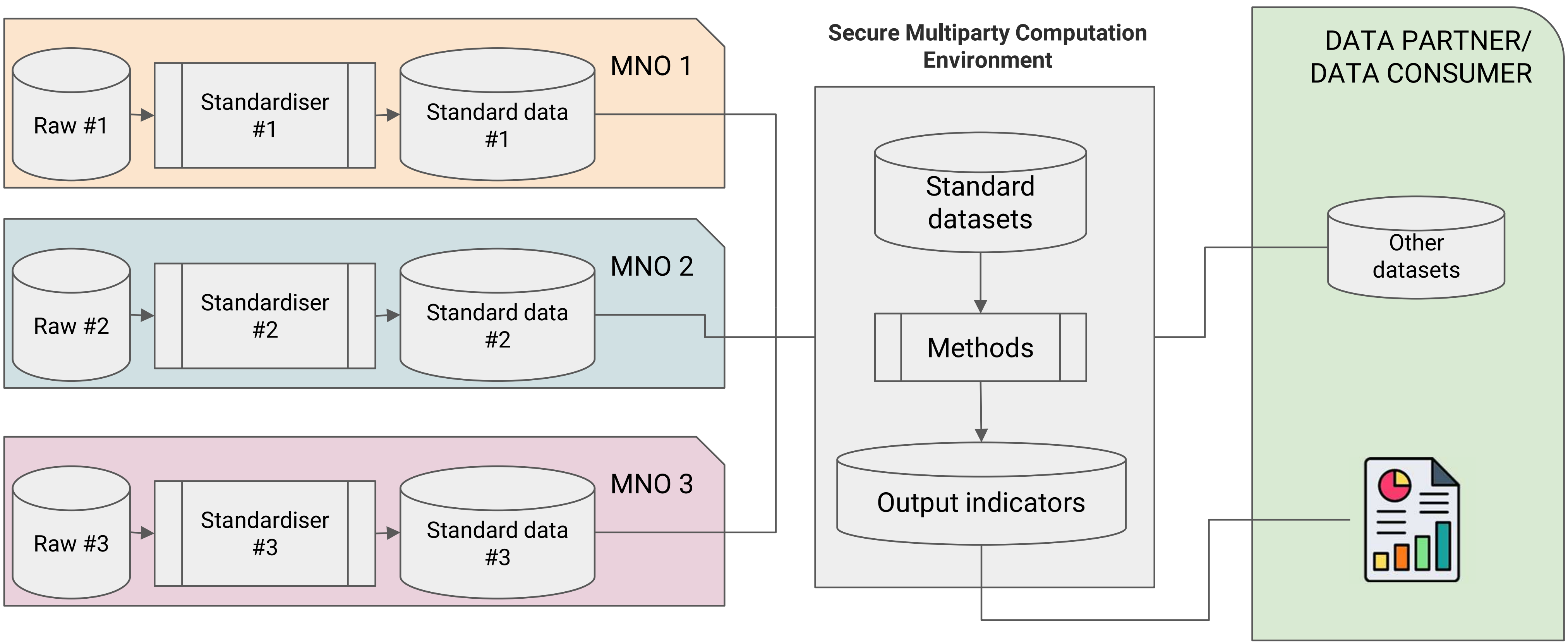
Strengths:

- Strong privacy guarantees
- Easy to scale across operators
- Legally and ethically compliant
- Requires modest infrastructure (in compare to other approaches)

Challenges and Limitations:

- Potential user duplication across MNOs
- Differences in data quality or methodological approach
- Loss of information due to privacy-protection techniques (e.g. k-anonymity)

MODEC approach - general schema



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MODEC approach

Phase 1: Secure Data Sharing Preparation

The MODEC approach enables multiple MNOs to jointly compute statistical indicators at the device level, while strictly preserving privacy. Unlike aggregate-based methods, MODEC operates on pseudonymized device-level data processed via Privacy-Enhancing Technologies (PETs).

Each MNO prepares its data as follows:

- **Raw Data Filtering**: Selecting relevant device events according to the indicator of interest (e.g., presence, mobility flows, dwell time).
- **Pseudonymization**: Device identifiers are irreversibly hashed or transformed to prevent tracing individuals.
- **PET-Compatible Packaging**: Data is formatted and encrypted to enable secure multiparty computation without revealing raw inputs.

MODEC approach

Phase 1: Secure Data Sharing Preparation

MNOs do not share raw or direct device data openly; instead, they participate in **cryptographic protocols** such as:

- Secure Multiparty Computation (SMPC)
- Private Set Intersection (PSI)
- Federated Analytics

IMPORTANT: All data processing occurs within secure, encrypted environments, ensuring that no participant gains access to another's raw device-level data.

MODEC approach

Phase 2: Privacy-Preserving Joint Computation and Harmonized Indicators

Once each MNO transforms its data into a privacy-compliant format, the MODEC process continues with a joint, secure computation of indicators using advanced PETs. This phase ensures methodological consistency and precise integration of data across operators.

- **Unified Calculation Methodology**: Unlike MOAC, where each MNO may apply its own processing methods, MODEC performs all computations in a centralized virtual environment, using a single, harmonized algorithm across all datasets. This guarantees that the resulting indicators are methodologically consistent and fully comparable.
- **Cross-MNO Deduplication – Two Modes**:
  - **Basic Deduplication**: Devices seen in multiple MNOs are counted only once, ensuring no inflation in user-based indicators like population estimates.
  - **Enriched Deduplication (Advanced PETs)**: When supported by the protocol, data associated with the same device from different MNOs can be combined securely, enabling deeper insights without violating privacy.

MODEC approach

Phase 2: Privacy-Preserving Joint Computation and Harmonized Indicators

- **Single Privacy Filtering at Integration Level**: Unlike in MOAC (where each MNO applies its own privacy thresholds independently), MODEC applies privacy-preserving thresholds after combining the full dataset. This means that thresholds such as k-anonymity or noise addition are applied only once, on the full joint dataset, minimizing information loss and increasing data utility.

- **Aggregated Outputs Only**: The final output is a set of fully anonymized, aggregate indicators, such as population counts, mobility flows, or usage metrics, ready to be used for statistical or planning purposes.

MODEC approach

Strengths

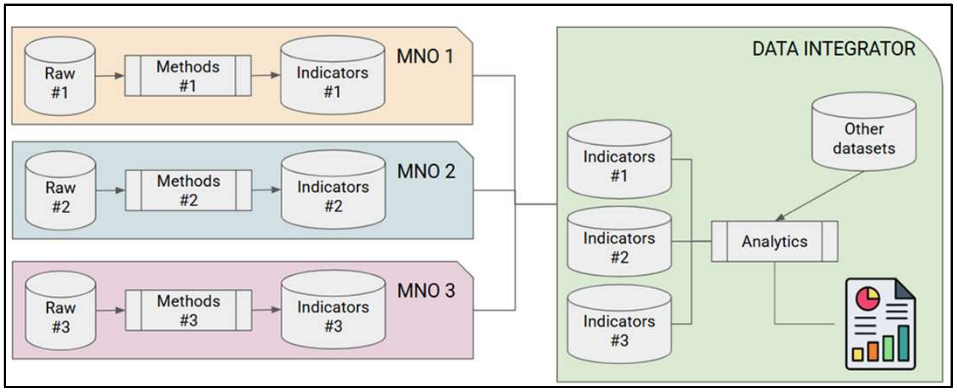
- Ensures **unique-user counting** across MNOs
- Guarantees **methodological uniformity**
- Enables **advanced indicators** through joint data usage (when technically allowed)
- **Privacy filtering applied once**, reducing unnecessary data suppression
- Strong **compliance** with data protection laws

Challenges

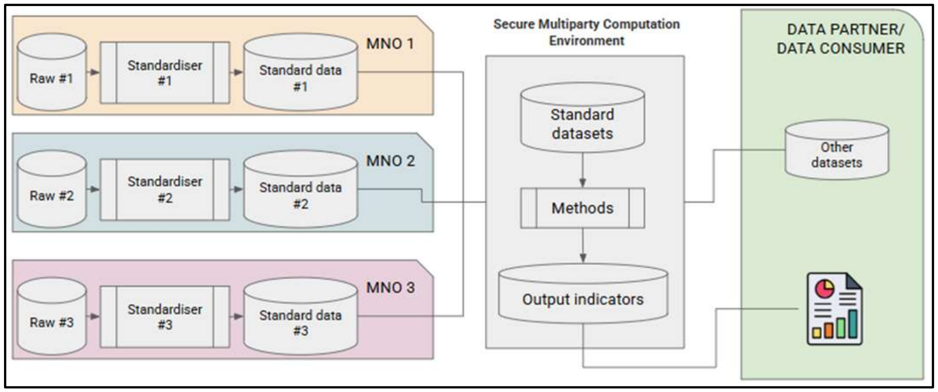
- Requires **legal agreements** and trust between MNOs
- High **technical complexity**
- May incur **greater costs** and delays than aggregate methods

Overview: MOAC vs MODEC

MOAC



MODEC



MOAC is simpler, faster, and often sufficient for high-level indicators.

MODEC unlocks richer, more accurate insights, at the cost of greater complexity

Overview: MOAC vs MODEC

Dimension	MOAC – Multi-Operator Aggregate-level Combination	MODEC – Multi-Operator Device-level Combination
Data Shared by MNOs	Aggregated indicators (e.g., total devices per area/hour)	Pseudonymized/encrypted device-level data (no raw identifiers)
Computation Location	Each MNO computes indicators independently	Joint computation in secure, privacy-preserving environment
Deduplication Across MNOs	❌ Not possible – devices may be counted more than once	✅ Possible – devices seen in multiple MNOs counted only once
Privacy Filtering	Applied independently by each MNO before sharing	Applied once, after integration – improves data utility
Methodological Consistency	⚠️ May differ across MNOs if methods vary	✅ Guaranteed – all data processed under a single, unified method

Overview: MOAC vs MODEC

Dimension	MOAC – Multi-Operator Aggregate-level Combination	MODEC – Multi-Operator Device-level Combination
Technical Complexity	Low to Medium	High – requires PETs (e.g., SMPC, PSI) and secure coordination
Privacy Risk	✅ Very low – only anonymized aggregates are shared	✅ Very low – PETs ensure no exposure of raw or personal data
Setup Requirements	Coordination of output format and privacy filters	⚠️ Legal, cryptographic, and organizational agreements between MNOs
Cost and Time	Lower setup, faster implementation	⚠️ Higher resource needs (time, expertise, infra)

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IV. Multi-MNO initiatives

Multi-MNO initiatives

In recent years, several projects and initiatives have explored how to combine data from MNOs to produce high-quality, privacy-compliant statistical indicators. These efforts respond to the growing need for more representative and accurate data products, especially for official statistics, mobility analysis, and population monitoring; by addressing the challenges of data heterogeneity, methodological consistency, and legal constraints. A selection of key initiatives that exemplify different approaches to the Multi-MNO paradigm, ranging from statistical aggregation to the application of advanced Privacy-Enhancing Technologies (PETs) for secure joint computation is presented in the following slides.

Note: The initiatives presented share a vision of integrating data from multiple MNOs; however, not all of them have multi-MNO data integration as their primary activity.

Multi-MNO initiatives

Eurostat ESSnet Big Data I & II (2016–2020)

Two large pan-European projects that explored "new data sources" for official stats, including extensive work on mobile operator data. Big Data I (2016–18) and II (2018–20) involved 22–28 NSIs (National Statistics Institutes) each and contained dedicated work-packages on mobile networks (e.g. "Mobile phone data" in Big Data I and "Mobile networks data" in Big Data II). These projects produced methods, case studies and public deliverables on using aggregated mobile data (from single and multiple operators) in domains like tourism and mobility.

https://cros.ec.europa.eu/group/cros-archives/book-page/past-projects

Multi-MNO initiatives

GSMA Big Data for Social Good (2017–present)

A global collaboration of mobile operators (initially 16 operator groups, 2+ billion subscribers) launched by GSMA. Participating MNOs standardize and share aggregated, anonymized mobile data indicators in a common format (subject to a strict privacy "code of conduct"). In pilots (e.g. Bharti Airtel in India, Telefónica in Brazil, Telenor in South Asia) the operators jointly produced mobility and calling statistics for governments/NGOs. The initiative focuses on public policy use cases (disaster response, health, social development) and emphasizes privacy: operators process pseudonymized data and only share high-level aggregates.

https://www.gsma.com/solutions-and-impact/connectivity-for-good/external-affairs/gsma_resources/big-data-toolkit/

Multi-MNO initiatives

ESS Task Force on MNO Data for Official Statistics (2021–present)

A Eurostat-led initiative bringing together 18 European national statistical institutes to define a common methodological framework for using mobile network operator (MNO) data in official statistics. The task force promotes multi-MNO integration, standardization of processing methods, and privacy-by-design principles. Its key goal is to produce harmonized, high-quality mobility and population indicators that are comparable across countries and operators. Outputs include public guidance documents, pilot implementations, and alignment with Eurostat's modernization agenda.

https://ec.europa.eu/eurostat/web/products-statistical-reports/w/ks-ft-23-001

Multi-MNO initiatives

UN Global Pulse – "DISHA" Data Insights (2021–present)

A UN-led multi-stakeholder initiative (UN Global Pulse, UNDP, WFP, Google.org, etc.) for ethical data and AI in humanitarian action. Among its projects, the Socio-Economic Mapping solution uses multi-operator mobile phone data to monitor population density, mobility and poverty in near-real time. For example, Disha's mapping nowcasts post-crisis population shifts to help target food and cash assistance. The initiative stresses standardized, privacy-preserving methods and has produced open tools/models for mobile data analysis.

https://disha.unglobalpulse.org/

Multi-MNO initiatives

Multi-MNO Project (2023–2025)

A cornerstone Eurostat-led project that defines an open, end-to-end methodological standard for processing raw data from multiple MNOs into aggregated statistical indicators. It includes the development of an open-source reference implementation and real-world testing across several EU countries. The project addresses the full pipeline, from raw data transformation to harmonised output, enabling comparable, privacy-aware multi-operator statistics https://cros.ec.europa.eu/landing-page/multi-mno-project

MNO-MINDS (2023–2025)

A European Statistical System (ESS)-funded project to develop methodologies and open-source tools for integrating mobile network data with other data sources. It emphasizes cross-operator calibration, survey-based validation, and includes work packages focused on Multi-MNO data harmonization and error assessment. https://cros.ec.europa.eu/mno-minds

Multi-MNO initiatives

PET4OS – Privacy-Enhancing Technologies for Official Statistics (2023–ongoing)

PET4OS is a European initiative coordinated by Eurostat that promotes the use of Privacy-Enhancing Technologies (PETs) in the production of official statistics. It supports a set of pilot projects aiming to enable secure, joint computation over sensitive microdata, such as mobile network operator (MNO) data, without requiring raw data sharing. Techniques like Secure Multiparty Computation (SMPC), Trusted Execution Environments (TEEs), and PET-as-a-Service platforms are tested across various use cases. PET4OS contributes to building a secure shared infrastructure to make cross-institutional data processing both privacy-preserving and legally compliant. Enables joint analysis of MNO data from different operators by applying PETs to compute integrated indicators (e.g., user counts, presence) without disclosing raw data between MNOs or to third parties.

https://cros.ec.europa.eu/PET4OS

V. Session exercise

Estimating Present Population Using Multi-MNO Mobile Network Data

You are part of a national statistics office exploring how to use mobile network data to estimate the **present population** in two cities of Portugal (Lisbon and Porto) at a given moment (10 A.M.). Mobile Network Operators (MNOs) are key data providers, but each operator only sees a portion of the population.

To improve accuracy, you are evaluating two Multi-MNO approaches:

- **MOAC** – Each MNO shares aggregate data
- **MODEC** – MNOs collaborate via privacy-preserving device-level computation

Your task is to compare the estimates for each city obtained using each approach, including the three single-MNO approaches, as well as the MOAC and MODEC methods. Discuss the possible reasons behind these differences and what they reveal about the strengths and limitations of each method.

Estimating Present Population Using Multi-MNO Mobile Network Data

TABLE 1 - Device identifiers per MNO at the regions under analysis at 10:00 AM

MNO / Devices identified	Zone 'Porto'	Zone 'Lisbon'
MNO 1 (50% of Market share)	D1, D2, D3, D4	D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21
MNO 2 (30% of Market share)	D6, D7, D1	D11, D12, D13, D14, D22, D23, D24, D25, D26, D27, D28, D29
MNO3 (20% of Market share)	D8, D9, D10, D2	D22, D23, D24, D25, D30, D31, D32

Note: Assume that all MNOs have agreed not to share any data points representing fewer than 10 devices, in line with a *k-anonymity threshold* of 10. This rule applies both when each MNO shares indicators individually (e.g., in MOAC), and when data is jointly processed under a MODEC scheme using privacy-preserving techniques.

Annex I. Exercise answer

Estimating Present Population Using Multi-MNO Mobile Network Data

IMPORTANT NOTE: The answer provided aims to present, in a general way, the main conceptual differences between single-MNO and multi-MNO (MOAC and MODEC) approaches. The methods shown in this example for estimating population from device counts, as well as the approaches for merging data from different MNOs, are simplified. The primary goal is to help understand the differences between the approaches rather than to deliver a sophisticated estimation or data merging method. For example, considerations such as differences in market share across territories, variations in social profiles, or the number of devices per person (e.g., some individuals having multiple devices while others, like children, may have none) are not taken into account in this example.

Estimating Present Population Using Multi-MNO Mobile Network Data

MNO	Zone 'Porto'	Devices per MNO	Information provided after k-anonimity	Estimation of people in Porto following Single MNO approach
MNO 1	D1, D2, D3, D4	4	'<10'	Considering that devices can vary between 1 and 9 and considering market share, the present population is between **2 and 18** (1/0.5 and 9/0.5)
MNO 2	D6, D7, D1	3	'<10'	Considering that devices can vary between 1 and 9 and considering market share, the present population is between **3 and 30** (1/0.3 and 9/0.3)
MNO3	D8, D9, D10, D2	4	'<10'	Considering that devices can vary between 1 and 9 and considering market share, the present population is between **5 and 45** (1/0.2 and 9/0.2)

Estimating Present Population Using Multi-MNO Mobile Network Data

MNO	Zone 'Porto'	Devices based on MOAC	Information provided after k-anonimity	Estimation of people in Porto based on MOAC
MNO 1	D1, D2, D3, D4	4	'<10'	All the MNOs report less than 10 devices, no providing exact information for privacy issues. Considering that devices are between 1 and 9 per MNO, and considering the market share of each operator, it can be estimated that present population in this region is between **3 and 31 people** ((1/0.5+1/0.3+1/0.2)/3=3.44 and (9/0.5+9/0.3+9/0.2)/3=31)
MNO 2	D6, D7, D1	3	'<10'	
MNO3	D8, D9, D10, D2	4	'<10'	

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Estimating Present Population Using Multi-MNO Mobile Network Data

<table>
<tr><th>MNO</th><th>Zone ‘Porto’</th><th>Devices based on MODEC</th><th>Information provided after k-anonimity</th><th>Estimation of people in Porto based on MODEC</th></tr>
<tr><td>MNO 1</td><td>D1, D2, D3, D4</td><td rowspan="3">9</td><td rowspan="3">‘<10’</td><td rowspan="3">It is reported less than 10 devices, no providing exact information for privacy issues. Considering that devices are between 1 and 9, and taking into account that operators account for the 100% of the market, it can be estimated that present population in this region is between 1 and 9 people</td></tr>
<tr><td>MNO 2</td><td>D6, D7, D1</td></tr>
<tr><td>MNO3</td><td>D8, D9, D10, D2</td></tr>
</table>

* Duplicates

Estimating Present Population Using Multi-MNO Mobile Network Data

MNO	Zone 'Lisbon'	Devices per MNO	Information provided after k-anonimity	Estimation of people in Porto following <u>Single MNO approach</u>
MNO 1	D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21	11	11	Estimation of present population is **22** (11/0.5)
MNO 2	D11, D12, D13, D14, D22, D23, D24, D25, D26, D27, D28, D29	12	12	Estimation of present population is **40** (12/0.3)
MNO3	D22, D23, D24, D25, D30, D31, D32	7	7	Estimation of present population is **35** (11/0.2)

Estimating Present Population Using Multi-MNO Mobile Network Data

MNO	Zone 'Lisbon'	Devices based on MOAC	Information provided after k-anonimity	Estimation of people in Porto based on MOAC
MNO 1	D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21	11	11	Estimation of number of present population: (11/0.5+12/0.3+7/0.2)/3 = **32.3**
MNO 2	D11, D12, D13, D14, D22, D23, D24, D25, D26, D27, D28, D29	12	12	
MNO3	D22, D23, D24, D25, D30, D31, D32	7	7	

Estimating Present Population Using Multi-MNO Mobile Network Data

MNO	Zone 'Lisbon'	Devices based on MODEC	Information provided after k-anonimity	Estimation of people in Porto based on MODEC
MNO 1	D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21	22	22	Estimation of number of present population: **22**
MNO 2	D11, D12, D13, D14, D22, D23, D24, D25, D26, D27, D28, D29			
MNO3	D22, D23, D24, D25, D30, D31, D32			

* Duplicates

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Estimating Present Population Using Multi-MNO Mobile Network Data

<table>
<tr><th>Approach</th><th>Zone ‘Porto’ results</th><th>Comments</th></tr>
<tr><td>Single-MNO 1</td><td>2-18</td><td rowspan="5">In the case of Porto, the sample of detected devices is very small, fewer than 10 unique devices when combining data from all MNOs, so no detailed information can be published under either the single-MNO or multi-MNO approach due to the k-anonymity threshold. As a result, population estimates must be derived based on plausible minimum and maximum values to define a possible range for the present population.

For the single-MNO approach, the range of uncertainty depends on the market share: the larger the market share, the narrower the estimation range. For instance, MNO1 (with 50% market share) yields a range between 2 and 18, whereas MNO3 (with only 20% share) results in a much wider interval of 5 to 45.

The most constrained and reliable estimate comes from the MODEC approach, which removes duplicates and accounts for 100% of the market, producing a narrow and privacy-compliant estimation range of 1 to 9.</td></tr>
<tr><td>Single-MNO 2</td><td>3-30</td></tr>
<tr><td>Single-MNO3</td><td>5-45</td></tr>
<tr><td>MOAC</td><td>3-31</td></tr>
<tr><td>MODEC</td><td>1-9</td></tr>
</table>

Estimating Present Population Using Multi-MNO Mobile Network Data

Approach	Zone 'Lisbon' results	Comments
Single-MNO 1	22	In the case of **Lisbon**, the number of detected devices is sufficiently high to allow each MNO to provide disaggregated population estimates without breaching privacy thresholds. This enables a meaningful comparison across methodological approaches. The individual estimates vary: MNO1 reports 22, MNO2 40, and MNO3 35. These differences may reflect natural variations in market share, user demographics, and data characteristics. While each single-MNO estimate is valid within its scope, combining multiple sources can improve representativeness. The **MOAC** approach integrates anonymized aggregate data from all MNOs and produces a combined estimate of 32.5. This enhances coverage but may include duplicated users. The **MODEC** approach, by contrast, uses joint privacy-preserving computation to deduplicate and harmonize across networks, yielding a final estimate of 22. Interestingly, this matches MNO1's result, not by design, but as a reflection of accurate deduplication and consistent methodology. While MODEC offers higher precision, single-MNO and MOAC approaches remain valuable tools, depending on feasibility and data availability.
Single-MNO 2	40	
Single-MNO3	35	
MOAC	32.5	
MODEC	22	

Annex II. Further Reading

Further Reading

[1] European Statistical System Task Force on the Use of Mobile Network Operator Data for Official Statistics. (2023). *Reusing mobile network operator data for official statistics: The case for a common methodological framework for the European Statistical System* (Statistical Reports No. KS-FT-23-001). Eurostat. https://ec.europa.eu/eurostat/documents/7870049/17468840/KS-FT-23-001-EN-N.pdf

[2] Ricciato, F.. (2023, December 8). *Reusing mobile network operator data for official statistics: The case for a common methodological framework for the European Statistical System* (ESS TF MNO Position Paper). Eurostat. https://cros.ec.europa.eu/system/files/2023-12/20231109_dimeitdg_mnoposition_paper_ricciato_ho%20%281%29.pdf

[3] Ricciato, F. (2024). *Steps toward a shared infrastructure for multi-party secure private computing in official statistics*. *Journal of Official Statistics, 40*(1), 3–15. https://doi.org/10.1177/0282423X241235259

[4] Coudin, E., Poulhes, M., & Suarez Castillo, M. (2021). The French official statistics strategy: Combining signaling data from various mobile network operators for documenting COVID-19 crisis effects on population movements and economic outlook. *Data & Policy*, *3*, E14. https://www.cambridge.org/core/journals/data-and-policy/article/french-official-statistics-strategy-combining-signaling-data-from-various-mobile-network-operators-for-documenting-covid19-crisis-effects-on-population-movements-and-economic-outlook/DBE5680E9B48AD780F11495D7255E91C

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